

DASITH EDIRISINGHE

+61475134104 | udasith@gmail.com | Brisbane, Australia

 Scholar |  ORCID |  LinkedIn |  GitHub |  Web

RESEARCH INTERESTS

Computer Vision, Representation Learning, Self-Supervision, Visual Reasoning

EDUCATION

Queensland University of Technology (QUT), Australia

Jun 2026 – Present

Doctor of Philosophy (PhD) - Computer Vision

University of Moratuwa, Sri Lanka

Nov 2018 – July 2023

BSc Engineering (Hons) in Computer Science and Engineering

- CGPA: 3.83 (First Class Honours)

PUBLICATIONS

- **Edirisinghe, D.**, Nimalsiri, W., Hennayake, M., Meedeniya, D., & Lim, G., "Chest X-Ray Report Generation Using Abnormality Guided Vision Language Model," in IEEE Access, vol. 13, pp. 157651-157673, 2025, doi: 10.1109/ACCESS.2025.3606961
- **Edirisinghe, D.**, Rajapakse, K., Abeysinghe, P., & Rathnayake, S. (2024). "SpotKube: Cost-Optimal Microservices Deployment with Cluster Autoscaling and Spot Pricing". In Proceedings of the 2024 IEEE International Conference on Cloud Computing Technology and Science (CloudCom), pp. 87–94. IEEE. DOI: 10.1109/Cloud-Com62794.2024.00026. (*Best Paper Nomination*)
- Herath, M., Mallawarachchi, A., Hasaranga, N., Gothatuwa, H., **Edirisinghe, D.**, Ambegoda, T. "Diffusion-Guided 3D Object Placement via 2D-to-3D Visual Consistency Optimization". Machine Vision and Applications - Manuscript in preparation.

Conference Presentations

- **Oral Presentation** - SpotKube: Cost-Optimal Microservices Deployment with Cluster Autoscaling and Spot Pricing - **Edirisinghe, D.**, Rajapakse, K., Abeysinghe, P., & Rathnayake, S. IEEE International Conference on Cloud Computing Technology and Science (CloudCom), Abu Dhabi, UAE, Dec 2024

WORK EXPERIENCE

Research Assistant - Computer Vision

March 2024 - May 2026

Artificial Intelligence Translational Research Lab (AITR), Sri Lanka - Onsite (Part Time)

Advisors: Dr. Thanuja D. Ambegoda, Prof. Dulani Meedeniya

- META-CXR: Method for chest radiograph understanding and report generation based on combined vision encoders with various inductive biases. (Accepted in IEEE Access) — 
- DG-3DPlace: Diffusion-guided 3D asset placement in 3D scenes using Gaussian Splatting
- TSR-GS: Self-supervised framework for transparent surface identification via 2D Gaussian optimization.
- ScholarMind: Exploring agentic AI frameworks to enhance research problem formulation and iterative reasoning with literature-grounded knowledge.

Machine Learning Engineer

July 2023 - Feb 2026

Rabot Inc, USA - Remote (Full Time)

- Designed and deployed real-time computer vision pipelines for automated pack stations, enabling multi-object detection and tracking at 10+ FPS under edge compute constraints
- Engineered a lightweight person-removal algorithm using prior-frame caching and detection-based masking; achieved real-time performance(20 ms per frame) on CPU-only inference, enhancing privacy compliance.

- Fine-tuned vision-language models (VLMs) for packaging-type recommendation, improving binary decision accuracy from 75% → 95% through targeted instruction tuning on A100 GPUs.
- Developed camera calibration and exposure control modules, combining a histogram-based auto-exposure algorithm and AprilTag-guided homography correction to stabilize detections under dynamic lighting and camera shifts.

Google Summer of Code Contributor

May 2022 - Sep 2022

Weaviate, Netherlands - Remote (Part Time)

Final Report 

- Developed the text summarization module for Weaviate, which helps users to summarize search results.
- Created the inference engine using Docker, FastAPI and the Hugging Face Transformers library.

Software Engineering Intern

Dec 2021 - Sep 2022

Sysco LABs, Sri Lanka - Onsite (Full Time)

- Fixed UI-related bugs using React JS and Improved the service layer performance by re-implementing logic and optimizing queries using Java and PLSQL

Software Engineering Intern

April 2021 - Sep 2021

LiveRoom Technologies, Sri Lanka - Remote (Part Time)

- Developed a 3D image conversion system leveraging AWS lambda container image support. [Project Link](#) 

HONORS AND AWARDS

QUT Postgraduate Research Award (QUTPRA): Awarded the prestigious fully funded QUTPRA scholarship for PhD studies at Queensland University of Technology (QUT), School of Electrical Engineering and Robotics (2026)

UTS President's Scholarship (UTSP): Awarded the prestigious President's Scholarship for PhD studies at the University of Technology Sydney (UTS), Faculty of Engineering and Information Technology (2026)

Best Paper Nomination (Top 5): IEEE International Conference on Cloud Computing Technology and Science (CloudCom), 2024. For the paper: "SpotKube: Cost-Optimal Microservices Deployment with Cluster Autoscaling and Spot Pricing."

Dean's List: Awarded for outstanding academic performance in Semesters 4, 5, and 8 at the Faculty of Engineering, University of Moratuwa.

Google Summer of Code 2022: Selected by Google for its highly competitive open-source program. Contributed to Weaviate and received a stipend and mentorship for the project: "Make a New Weaviate Module."

TEACHING EXPERIENCE

Teaching Assistant, CS3042: **Database Systems**, University of Moratuwa, Sri Lanka

Aug 2022 - Dec 2022

Teaching Assistant, CS3953: **Technical Writing**, University of Moratuwa, Sri Lanka

Aug 2022 - Dec 2022

TALKS

Google Summer of Code Awareness Session - CS&ES, University of Moratuwa, Sri Lanka

February 2023

Software Engineering Best Practices - CSE, University of Moratuwa, Sri Lanka

September 2022

Deep Learning based Recommendation Systems - LiveRoom Tech Talks

February 2021

REFERENCE

Prof. Dulani Meedeniya
Professor in Computer Science & Engineering
University of Moratuwa, Sri Lanka
+94 71 393 5801
dulanim@cse.mrt.ac.lk

Dr. Thanuja D. Ambegoda
Senior Lecturer in Computer Science & Engineering
University of Moratuwa, Sri Lanka
+94 70 265 5273
thanuja@cse.mrt.ac.lk